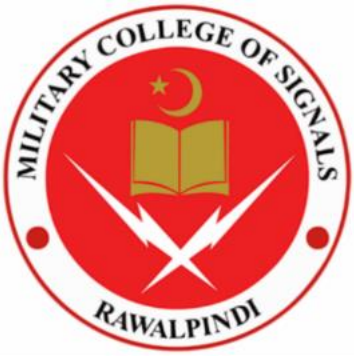




MILITARY COLLEGE OF SIGNALS

DEPARTMENT OF COMPUTER SOFTWARE ENGINEERING

Assignment - 1

Fundamentals of Computer Programming	
Submitted To	Prof. Dr. Imran Usman
Submitted By	Muhammad Rehan
Class	32 (B)

Printing stars using C++.

Task 1:

Source code:

```
#include <iostream>
using namespace std;

int main()
{

int n;
cout << "Enter number of lines of stars you want:\n";
cin >> n;

for (int i = 0; i < n; i++)
{
for (int j = 0; j < (i+1); j++)
{
cout << "*";
}
for(int k = i; k < (n*2)-(i+1); k++)
{
cout << " ";
}
for (int l = 0; l < (i+1); l++)
{
cout << "*";
}
cout << "\n";
}
return 0;
}
```

star10.cpp > ...

```cpp
#include <iostream>
using namespace std;

int main()
{

    int n;
    cout << "Enter number of lines of stars you want:\n";
    cin >> n;

    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < (i+1); j++)
        {
            cout << "*";
        }
        for(int k = i; k < (n*2)-i; k++)
        {
            cout << " ";
        }
        for (int l = 0; l < (i+1); l++)
        {
            cout << "*";
        }
        cout << "\n";

    }

    return 0;
}




/*
*       *
**     **
***   ***
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
rehanshahid@Muhammads-MacBook-Air Tasks % ./star10
Enter number of lines of stars you want:
4
*       *
**     **
***   ***
**** ****
rehanshahid@Muhammads-MacBook-Air Tasks %
```

Taks2:

Source code:

```
#include <iostream>
using namespace std;

int main()
{
int n;

cout << "Enter number of lines you want star pattern:\n";
cin >> n;

for(int i = 0; i < n-1; i++)
{
for(int j = 0; j < (n-(i+1)); j++)
{
cout << " ";
}
for(int k = 0; k < (i+1); k++)
{
cout << "* ";
}
cout << "\n";
}

//Inverted pattern for completing tree
for(int i = 0; i < n; i++)
{
for(int l = 0; l < i; l++)
{
cout << " ";
}
for(int m = 0; m < (n-i); m++)
{
cout << "* ";
}
cout << "\n";
}

return 0;
}
```

star11.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;

    cout << "Enter number of lines you want star pattern:\n";
    cin >> n;

    for(int i = 0; i < n-1; i++)
    {
        for(int j = 0; j < (n-(i+1)); j++)
        {
            cout << " ";
        }
        for(int k = 0; k < (i+1); k++)
        {
            cout << "* ";
        }
        cout << "\n";
    }

    //Inverted pattern for completing tree
    for(int i = 0; i < n; i++)
    {
        for(int l = 0; l < i; l++)
        {
            cout << " ";
        }
        for(int m = 0; m < (n-i); m++)
        {
            cout << "* ";
        }
        cout << "\n";
    }

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
rehanshahid@Muhammads-MacBook-Air Tasks % ./star11
Enter number of lines you want star pattern:
4
   *
  * *
 * * *
* * * *
 * * *
  * *
   *
```